Cloud Computing – Question Bank

NPTEL, IIT Kharagpur. Each answer shows the correct option, why it is right, and why every other option is wrong.

Week 1

Week 1 · Question 1

Which of the following best represents the meaning of resource pooling in cloud computing?

A. Every customer receives a permanently dedicated physical server.

B. Computing resources are dynamically assigned to multiple customers using a shared infrastructure.

C. Resources can only be accessed from a local network.

D. Users must manually allocate hardware before using cloud services.

Correct answer: B

Why B is right: Resource pooling is a NIST essential characteristic: one shared pool of resources serves many customers and is assigned and reassigned as demand changes.

Why the other options are wrong

A is wrong: A dedicated server per customer is the opposite of pooling.

C is wrong: Cloud resources are reachable over the network (broad network access), not only a local one.

D is wrong: Cloud users provision resources on demand with no manual hardware allocation.

Week 1 · Question 2

A startup rents virtual servers on Amazon EC2. The team installs and patches its own operating system, configures the web server, and manages the runtime and applications itself. The provider supplies only the virtualized compute, storage, and networking. This is best classified as:

A. SaaS

B. PaaS

C. IaaS

D. FaaS

Correct answer: C

Why C is right: In IaaS the provider gives virtual compute, storage and networking, and the customer manages the OS, runtime and applications.

Why the other options are wrong

A is wrong: In SaaS the provider runs everything, including the application.

B is wrong: In PaaS the provider also manages the OS and runtime; the customer only deploys code.

D is wrong: In FaaS the customer uploads small functions and manages no servers or OS.

Week 1 · Question 3

Which deployment model allows an organization to maintain cloud infrastructure exclusively for its own use, either on-premises or hosted by a third party?

A. Public Cloud

B. Hybrid Cloud

C. Community Cloud

D. Private Cloud

Correct answer: D

Why D is right: A private cloud is dedicated to one organization, giving more control over security, compliance and resources.

Why the other options are wrong

A is wrong: A public cloud is open to the general public and shared.

B is wrong: A hybrid cloud combines private and public clouds, so it is not exclusive.

C is wrong: A community cloud is shared by several organizations with common concerns.

Week 1 · Question 4

Virtualization technology allows multiple virtual machines (VMs) to run on a single physical system or on a cluster of physical systems. Which of the following statements is the MOST accurate?

A. Each VM always shares the same operating system kernel with the host OS.

B. Multiple VMs can run different operating systems on the same physical machine simultaneously.

C. Virtualization is only possible in cloud computing environments.

D. Virtualization prevents resource overcommitment on the host system.

Correct answer: B

Why B is right: Hypervisors such as VMware and KVM let VMs with different guest operating systems share the same hardware at once.

Why the other options are wrong

A is wrong: Sharing the host kernel describes containers; a full VM has its own guest kernel.

C is wrong: Virtualization existed before cloud computing and works on ordinary servers and desktops.

D is wrong: Overcommitting CPU or memory is a commonly used feature, not something virtualization prevents.

Week 1 · Question 5

Consider the following statements about distributed systems (DS): (i) DS can improve scalability by adding more nodes to the system. (ii) DS can increase fault tolerance through data and service replication. (iii) DS guarantees lower latency for every single operation compared to a centralized system. Choose the MOST APPROPRIATE option:

A. (i) and (ii)

B. Only (iii)

C. (i) and (iii)

D. (i), (ii), and (iii)

Correct answer: A

Why A is right: (i) and (ii) are true. (iii) is false because distribution adds network communication and coordination overhead, so some operations get slower.

Why the other options are wrong

B is wrong: Statement (iii) is false, so "only (iii)" cannot be right; it also drops the true (i) and (ii).

C is wrong: This includes the false statement (iii).

D is wrong: This also includes the false statement (iii).

Week 1 · Question 6

The combination of Service-Oriented Architecture (SOA) and Cloud Computing led to a paradigm where almost any IT capability can be delivered as a network-accessible service. This is often referred to as:

A. FTP

B. SNTP

C. XaaS

D. FaaS

Correct answer: C

Why C is right: XaaS means Anything/Everything as a Service, the umbrella term for delivering diverse IT capabilities as services.

Why the other options are wrong

A is wrong: FTP is a file transfer protocol.

B is wrong: SNTP is a time synchronization protocol.

D is wrong: FaaS (Functions as a Service) is only one specific service model, not the broad paradigm.

Week 1 · Question 7

Which of the following correctly describes a Type-2 (hosted) hypervisor?

A. It runs directly on the physical hardware without any underlying operating system.

B. It runs as an application on top of a host operating system.

C. It always provides lower virtualization overhead than a Type-1 hypervisor.

D. It cannot run more than one virtual machine at a time.

Correct answer: B

Why B is right: A Type-2 hypervisor (VirtualBox, VMware Workstation) is an application running on a host OS.

Why the other options are wrong

A is wrong: That describes a Type-1 (bare-metal) hypervisor.

C is wrong: Type-1 usually has lower overhead, because Type-2 goes through the host OS.

D is wrong: Hosted hypervisors can run many VMs at the same time.

Week 1 · Question 8

Which of the following are advantages of Platform-as-a-Service (PaaS)? (i) Developers can focus primarily on application development. (ii) Automatic software updates for the underlying platform. (iii) Complete control over physical server hardware. (iv) Faster application deployment.

A. (i), (ii), and (iv)

B. (ii) and (iii)

C. (i) and (iii)

D. (i), (ii), (iii), and (iv)

Correct answer: A

Why A is right: PaaS hides infrastructure and provides a managed runtime, so (i), (ii) and (iv) are advantages.

Why the other options are wrong

B is wrong: It includes (iii), which is false, and leaves out (i) and (iv).

C is wrong: It includes (iii), which is false, and leaves out (ii) and (iv).

D is wrong: It includes (iii): PaaS users never control physical hardware.

Week 1 · Question 9

Which of the following services is most likely to be classified as Infrastructure as a Service (IaaS)?

A. Microsoft Word Online

B. Amazon EC2 Virtual Machines

C. Google Docs

D. Salesforce CRM

Correct answer: B

Why B is right: EC2 provides virtual machines where users manage the OS, software and applications, which is IaaS.

Why the other options are wrong

A is wrong: Word Online is a ready-to-use application (SaaS).

C is wrong: Google Docs is a ready-to-use application (SaaS).

D is wrong: Salesforce CRM is a ready-to-use application (SaaS).

Week 1 · Question 10

A _____ is a form of distributed computing in which a large number of geographically dispersed and often heterogeneous resources across multiple administrative domains are loosely coupled and coordinated to work together on a common, large-scale task.

A. Cluster

B. Grid

C. Node

D. Hypervisor

Correct answer: B

Why B is right: A grid coordinates dispersed, heterogeneous, loosely coupled resources from several administrative domains.

Why the other options are wrong

A is wrong: A cluster is tightly coupled and usually located together.

C is wrong: A node is just one machine.

D is wrong: A hypervisor is virtualization software, not a distributed system.

Week 2

Week 2 · Question 1

A virtualization solution requires the guest operating system to invoke hypercalls instead of executing certain privileged instructions directly. This approach is known as:

A. Full Virtualization

B. Hardware-assisted Virtualization

C. Para-virtualization

D. Containerization

Correct answer: C

Why C is right: Para-virtualization modifies the guest OS so privileged operations go through hypercalls to the hypervisor. This lowers overhead but needs OS modification.

Why the other options are wrong

A is wrong: Full virtualization runs an unmodified guest OS, so no hypercalls are needed.

B is wrong: Hardware-assisted virtualization uses CPU extensions with an unmodified guest OS.

D is wrong: Containers share the host kernel and have no guest OS making hypercalls.

Week 2 · Question 2

Which of the following is a key characteristic of full (hardware-assisted) virtualization, as opposed to para-virtualization?

A. Guest OS must be modified to issue hypercalls to the hypervisor.

B. Guest OS runs unmodified, with the hypervisor and CPU extensions handling privileged instructions.

C. Guest and host must share the same operating system kernel.

D. No hypervisor is involved at all.

Correct answer: B

Why B is right: The guest OS stays unmodified while CPU extensions (Intel VT-x, AMD-V) and the hypervisor handle privileged instructions transparently.

Why the other options are wrong

A is wrong: That describes para-virtualization.

C is wrong: Sharing one kernel describes containers.

D is wrong: A hypervisor is always present in virtualization.

Week 2 · Question 3

Which statement about a public cloud is MOST accurate?

A. It is dedicated to, and controlled by, a single organization behind its own firewall.

B. Its infrastructure is provisioned for open use by the general public and is owned and operated by a cloud service provider.

C. It is shared only among organizations that have common regulatory concerns.

D. It always guarantees stronger data isolation than a private cloud.

Correct answer: B

Why B is right: A public cloud is open to the general public and is owned, managed and operated by a cloud service provider on its own premises.

Why the other options are wrong

A is wrong: That describes a private cloud.

C is wrong: That describes a community cloud.

D is wrong: Private clouds typically provide stronger isolation, and "always" makes this false.

Week 2 · Question 4

An application must repeatedly navigate, randomly access, and modify elements of a moderately sized XML document held in memory. Which XML parser API is MOST appropriate?

A. SAX (push-based, event-driven, forward-only)

B. StAX (pull-based, event-driven, forward-only)

C. DOM (builds an in-memory tree allowing random access and modification)

D. None; an XML document cannot be modified once parsed

Correct answer: C

Why C is right: DOM builds the whole document as an in-memory tree, so you can jump to any element and change it.

Why the other options are wrong

A is wrong: SAX is forward-only streaming; it cannot revisit or modify elements.

B is wrong: StAX is also forward-only streaming and not built for random access or edits.

D is wrong: XML documents can be modified once parsed, for example with DOM.

Week 2 · Question 5

Which statement correctly distinguishes a 'host OS', a 'guest OS', and a 'hypervisor'?

A. The host OS runs inside a VM, the guest OS runs on bare metal, and the hypervisor is a user application.

B. The hypervisor creates and manages VMs; each guest OS runs inside a VM; and for a Type-2 hypervisor, the host OS is the OS on which the hypervisor runs.

C. The guest OS and host OS are always identical copies of the same operating system.

D. The hypervisor is simply the hardware layer located beneath the host OS.

Correct answer: B

Why B is right: The hypervisor creates and manages VMs, each guest OS runs inside its own VM, and in a Type-2 setup the host OS is the OS the hypervisor runs on.

Why the other options are wrong

A is wrong: This reverses the host and guest roles.

C is wrong: Guest and host can be completely different operating systems.

D is wrong: The hypervisor is software, not a hardware layer.

Week 2 · Question 6

An enterprise migrates its private cloud to a third-party managed data center. Which of the following concerns should be evaluated? (i) Service availability guarantees (SLAs) (ii) Compliance with data protection regulations (iii) Network latency between users and cloud resources

A. Only (i)

B. (i) and (ii)

C. (ii) and (iii)

D. (i), (ii), and (iii)

Correct answer: D

Why D is right: All three matter: the applications now depend on an external provider's SLAs, legal rules about data location, and network performance.

Why the other options are wrong

A is wrong: Checking only SLAs ignores compliance and latency.

B is wrong: This ignores network latency.

C is wrong: This ignores the SLA guarantees.

Week 2 · Question 7

In the traditional web services stack, which technology is used to describe a web service's interface — its operations, input/output messages, and how to invoke it?

A. UDDI

B. WSDL

C. SMTP

D. HTML

Correct answer: B

Why B is right: WSDL is an XML-based language that describes a service's operations, message formats and binding details.

Why the other options are wrong

A is wrong: UDDI is the registry used to discover services, not to describe them.

C is wrong: SMTP is an email transport protocol.

D is wrong: HTML is a markup language for web pages.

Week 2 · Question 8

Traditional SOAP-based web services provide which of the following capabilities? (i) Platform-independent communication (ii) XML-based message exchange (iii) Automatic service discovery through UDDI (iv) Guaranteed microsecond-level response latency

A. (i), (ii), and (iii)

B. (ii) and (iv)

C. Only (i)

D. All of the above

Correct answer: A

Why A is right: SOAP services give interoperability using XML and WSDL, and discovery through UDDI. They cannot promise microsecond latency because XML processing adds overhead.

Why the other options are wrong

B is wrong: It includes (iv), which is false, and leaves out (i) and (iii).

C is wrong: It lists only (i), but (ii) and (iii) are also true.

D is wrong: It includes (iv), which is false.

Week 2 · Question 9

Which of the following problems are effectively addressed by traditional web services, and which are NOT? (i) Passing through firewalls using standard HTTP/HTTPS (ii) Enabling interoperability across different programming languages and operating systems (iii) Ultra-low-latency, real-time inter-service communication (iv) Service discovery through registries like UDDI

A. Only (i) and (ii) are addressed

B. (i), (ii), and (iv) are addressed; (iii) is NOT addressed

C. Only (ii) is addressed

D. All are addressed

Correct answer: B

Why B is right: HTTP/HTTPS, XML/SOAP/WSDL and UDDI cover (i), (ii) and (iv). XML parsing and SOAP overhead add delay, so (iii) is not addressed; gRPC suits real-time needs better.

Why the other options are wrong

A is wrong: This misses (iv), which UDDI does address.

C is wrong: This misses (i) and (iv), which are also addressed.

D is wrong: This wrongly claims (iii) is addressed.

Week 2 · Question 10

In the Service-Oriented Architecture (SOA) model, which set of operations correctly maps the interactions among the service provider, service registry, and service requestor?

A. Publish (provider → registry), Find (requestor → registry), Bind (requestor → provider)

B. Publish (requestor → provider), Find (provider → registry), Bind (registry → requestor)

C. Bind (provider → registry), Publish (requestor → registry), Find (provider → requestor)

D. Find (provider → registry), Publish (requestor → registry), Bind (provider → provider)

Correct answer: A

Why A is right: The provider publishes its description to the registry, the requestor finds it there, then binds to the provider and invokes it directly.

Why the other options are wrong

B is wrong: Here the roles are mixed up: the requestor does not publish to the provider.

C is wrong: Here the provider binds to the registry and the requestor publishes, which reverses their roles.

D is wrong: Here the requestor publishes and the provider binds to itself, which makes no sense.

Week 3

Week 3 · Question 1

Which of the following is least likely to be specified in an operational cloud Service Level Agreement (SLA)?

A. Recovery Time Objective (RTO)

B. Exact encryption algorithm and key storage procedures

C. Mean Time To Repair (MTTR) for incidents

D. Penalty / credit calculation for downtime

Correct answer: B

Why B is right: An SLA states service-level commitments. Low-level security details such as exact algorithms and key storage belong in security or operational documents.

Why the other options are wrong

A is wrong: RTO is a recovery commitment and is commonly stated in SLAs.

C is wrong: MTTR measures how fast incidents are fixed and is commonly stated.

D is wrong: Credits or penalties for downtime are a standard SLA term.

Week 3 · Question 2

A multilevel SLA generally consists of which of the following levels? (i) Corporate Level (ii) Customer Level (iii) Service Level

A. Only (i)

B. (i) and (ii)

C. (ii) and (iii)

D. (i), (ii), and (iii)

Correct answer: D

Why D is right: A multilevel SLA has corporate-level terms for all customers, customer-level terms for specific clients, and service-level terms for individual services.

Why the other options are wrong

A is wrong: This leaves out the customer and service levels.

B is wrong: This leaves out the service level.

C is wrong: This leaves out the corporate level.

Week 3 · Question 3

An enterprise signs a three-tier multilevel SLA with a cloud provider. However, the corporate-level SLA guarantees 95% uptime, the customer-level SLA specifies 98% uptime, and the service-level SLA guarantees 99.9% uptime. In case of a service disruption, which level of SLA is the provider LEGALLY BOUND to honor?

A. Corporate-level SLA, as it's the broadest agreement

B. Customer-level SLA, as it's the middle ground

C. Service-level SLA, as it's the most specific agreement

D. The provider can choose the lowest applicable SLA

Correct answer: C

Why C is right: As given in the course answer key: the service-level SLA is the most specific to actual service delivery, so accountability lies there (99.9%).

Why the other options are wrong

A is wrong: Being the broadest does not make the corporate level the binding one in this answer key.

B is wrong: Being the middle ground does not make the customer level the binding one.

D is wrong: The provider cannot choose the weakest commitment to escape its obligations.

Week 3 · Question 4

Which pair of metrics is most directly used to quantify the reliability and recoverability commitments in a cloud SLA?

A. Availability (uptime %) and Mean Time To Recovery (MTTR)

B. Lines of code and number of microservices

C. Number of employees and office locations

D. CPU clock speed and RAM size of the servers

Correct answer: A

Why A is right: Availability measures reliability and MTTR measures how quickly service is restored after an incident.

Why the other options are wrong

B is wrong: Code size and microservice count are implementation details, not SLA commitments.

C is wrong: Staff numbers and offices are organizational facts, not service metrics.

D is wrong: Hardware specifications are low-level details, not SLA commitments.

Week 3 · Question 5

A cloud service guarantees 99.9% availability per 30-day month. What is the maximum allowable downtime in that month under the SLA? (Assume 30 days = 720 hours.)

A. About 0.72 hours

B. About 7.2 hours

C. About 3.6 hours

D. About 72 minutes

Correct answer: A

Why A is right: Allowed downtime = (1 − 0.999) × 720 = 0.72 hours, about 43 minutes.

Why the other options are wrong

B is wrong: 7.2 hours is 1% of 720, which matches 99%, not 99.9%.

C is wrong: 3.6 hours is 0.5% of 720, which matches 99.5%.

D is wrong: 72 minutes is 1.2 hours, more than the 0.72 hours allowed.

Week 3 · Question 6

Which feature primarily enables a parallel database to process large analytical queries faster than a traditional database?

A. Exclusive use of SQL

B. Distributed query execution across multiple computing nodes

C. Elimination of indexing

D. Storing all data in a single server

Correct answer: B

Why B is right: A parallel database partitions data and runs query operations on many nodes at the same time.

Why the other options are wrong

A is wrong: SQL is only the query language; traditional databases use it too.

C is wrong: Removing indexing does not add parallelism.

D is wrong: One server is the traditional design, with no parallel speed-up.

Week 3 · Question 7

You are building a fraud-detection pipeline that must flag suspicious transactions within a few hundred milliseconds as they arrive. The system must: ingest about 500,000 transactions per second; detect and alert within sub-second latency; apply per-transaction rules and short sliding-window aggregations. Which architectural choice is LEAST suitable?

A. Apache Kafka feeding Apache Flink for continuous stream processing

B. Amazon Kinesis with a real-time stream-processing application

C. A nightly batch job on a data warehouse that scans the day's transactions

D. Apache Pulsar with a stream processor for windowed aggregation

Correct answer: C

Why C is right: A nightly batch job adds hours of delay and cannot give sub-second detection.

Why the other options are wrong

A is wrong: Kafka with Flink is built for low-latency continuous processing, so it suits the need.

B is wrong: Kinesis with a real-time application handles streaming at this scale, so it suits the need.

D is wrong: Pulsar with a stream processor supports windowed aggregation, so it suits the need.

Week 3 · Question 8

A company stores petabytes of satellite imagery that must be processed in parallel by Spark clusters. Which storage system is most suitable?

A. Local SSD attached to a single virtual machine

B. Object storage such as Amazon S3 or Google Cloud Storage

C. Relational database server

D. Client-side browser cache

Correct answer: B

Why B is right: Cloud object storage scales almost without limit, is durable, and works directly with Spark and Hadoop.

Why the other options are wrong

A is wrong: One VM's SSD is too small and cannot be shared across a cluster.

C is wrong: A relational server is not designed for petabytes of imagery.

D is wrong: A browser cache is temporary storage on a user's device.

Week 3 · Question 9

A MapReduce job is designed to count word frequencies in 2 TB of text files. The mapper output is 1.5 TB (including overhead), and after the shuffle and sort phase, the data is 1.3 TB. During the reduce phase, a node runs out of disk space during the merge operation. Which of the following is the most likely cause and solution?

A. The mapper logic is inefficient; rewrite the mapper function

B. The combiner phase didn't run; ensure combiner is explicitly configured to compress intermediate data locally

C. The shuffle and sort didn't deduplicate keys; increase the number of reducers

D. The network bandwidth is saturated; add more nodes to the cluster

Correct answer: B

Why B is right: A combiner pre-aggregates mapper output on each mapper node, shrinking the data before the shuffle so reducers do not run out of disk during the merge.

Why the other options are wrong

A is wrong: Mapper logic affects correctness, not the volume reaching the reducers' disks.

C is wrong: Shuffle and sort is not meant to deduplicate keys; reducers group them.

D is wrong: Saturated bandwidth slows transfer; it does not fill a node's disk.

Week 3 · Question 10

An OpenStack compute node contains 640 physical vCPUs. If the CPU allocation ratio is 5:1, what is the maximum number of virtual CPUs that may be allocated?

A. 3000

B. 3200

C. 3400

D. 3500

Correct answer: B

Why B is right: Allocatable vCPUs = 640 × 5 = 3200.

Why the other options are wrong

A is wrong: 3000 is 640 × 4.69, not a 5:1 ratio.

C is wrong: 3400 is more than 640 × 5.

D is wrong: 3500 is more than 640 × 5.

Week 4

Week 4 · Question 1

Which Azure SQL Database service tier separates compute and storage so that storage can automatically grow up to 100 TB without requiring application downtime?

A. Basic

B. Business Critical

C. Hyperscale

D. Standard

Correct answer: C

Why C is right: Hyperscale decouples compute from storage, so storage can scale to 100 TB, and it supports fast backup and restore.

Why the other options are wrong

A is wrong: Basic is a small entry tier with limited size.

B is wrong: Business Critical focuses on high availability and low latency, not 100 TB decoupled storage.

D is wrong: Standard is a general tier without decoupled 100 TB storage.

Week 4 · Question 2

One company deploys two Azure Virtual Machines in different Availability Zones within the same Azure region. One data center experiences a complete power outage. Which statement is correct?

A. Both VMs will become unavailable because they are in the same region.

B. The second VM can continue running because Availability Zones are isolated from each other.

C. Azure automatically moves the failed VM to another region.

D. Both VMs share the same cooling and networking infrastructure.

Correct answer: B

Why B is right: Availability Zones are separate data centers with independent power, cooling and networking, so the other zone keeps running.

Why the other options are wrong

A is wrong: Being in the same region does not mean sharing one failure.

C is wrong: Azure does not automatically move a failed VM to another region.

D is wrong: Zones do not share cooling or networking; that is why they exist.

Week 4 · Question 3

Which App Service pricing tier supports the highest number of deployment slots?

A. Basic

B. Standard

C. Premium (P1v3 and higher)

D. Free

Correct answer: C

Why C is right: Premium supports up to 20 deployment slots, suited to advanced DevOps and CI/CD.

Why the other options are wrong

A is wrong: Basic does not support deployment slots.

B is wrong: Standard supports far fewer slots than Premium.

D is wrong: Free does not support deployment slots.

Week 4 · Question 4

An administrator notices that users can still access an Azure App Service website using http://. Which App Service configuration should be enabled to prevent unencrypted access?

A. Managed Identity

B. Always On

C. HTTPS Only

D. Deployment Slots

Correct answer: C

Why C is right: HTTPS Only redirects every HTTP request to HTTPS, so traffic is encrypted.

Why the other options are wrong

A is wrong: Managed Identity lets the app authenticate to other Azure services.

B is wrong: Always On keeps the app loaded and warm.

D is wrong: Deployment slots are staging environments for releases.

Week 4 · Question 5

Which Google Cloud database service is designed specifically for structured relational data with global consistency and horizontal scalability?

A. Firestore

B. Cloud SQL

C. Cloud Spanner

D. Memorystore

Correct answer: C

Why C is right: Cloud Spanner is a globally distributed relational database with SQL, horizontal scaling and strong consistency.

Why the other options are wrong

A is wrong: Firestore is a NoSQL document database.

B is wrong: Cloud SQL scales mainly vertically and is not globally distributed.

D is wrong: Memorystore is an in-memory cache, not a relational database.

Week 4 · Question 6

A virtual machine requires an additional 100 GB persistent disk that should remain intact even after the VM is terminated. Which OpenStack service manages this storage?

A. Nova

B. Swift

C. Cinder

D. Keystone

Correct answer: C

Why C is right: Cinder provides persistent block storage that lives independently of the VM.

Why the other options are wrong

A is wrong: Nova manages compute instances, not persistent disks.

B is wrong: Swift is object storage, not a block disk attached to a VM.

D is wrong: Keystone handles identity and authentication.

Week 4 · Question 7

Which statement correctly distinguishes GKE from Cloud Run?

A. GKE only supports virtual machines.

B. Cloud Run requires users to manage Kubernetes clusters.

C. GKE manages Kubernetes clusters, while Cloud Run executes containers without requiring cluster management.

D. Both services require manual Kubernetes control plane management.

Correct answer: C

Why C is right: GKE is managed Kubernetes. Cloud Run is a serverless container platform where Google manages the infrastructure and scaling.

Why the other options are wrong

A is wrong: GKE runs containers on Kubernetes, not only VMs.

B is wrong: Cloud Run hides clusters; users do not manage Kubernetes.

D is wrong: In both services Google manages the control plane.

Week 4 · Question 8

A company collects millions of IoT sensor events every minute. The data must be ingested in real time, stored in a scalable analytical warehouse, and visualized through interactive dashboards. Which GCP architecture best satisfies these requirements?

A. Cloud SQL + Cloud Functions + Looker

B. Pub/Sub + BigQuery + Looker Studio

C. Firestore + Compute Engine + Bigtable

D. Cloud Storage + Memorystore + App Engine

Correct answer: B

Why B is right: Pub/Sub ingests events in real time, BigQuery stores and queries petabyte-scale analytical data, and Looker Studio builds dashboards on it.

Why the other options are wrong

A is wrong: Cloud SQL is not suited to high-volume analytics.

C is wrong: Firestore is operational NoSQL storage and this set has no analytical warehouse or dashboards.

D is wrong: Memorystore is a cache and this set has no warehouse or real-time ingestion service.

Week 4 · Question 9

A DevOps engineer wants to route all production traffic to version v3 of an App Engine service. Which command is correct?

A. gcloud app services set-traffic default --splits=v3=1

B. gcloud app deploy --traffic=v3

C. gcloud app versions promote v3

D. gcloud app services update v3

Correct answer: A

Why A is right: A traffic split of v3=1 sends 100% of traffic to version v3.

Why the other options are wrong

B is wrong: gcloud app deploy has no --traffic=v3 option for this.

C is wrong: This is not the command for setting a traffic split.

D is wrong: This command does not exist in this form.

Week 4 · Question 10

Which OpenStack service is primarily responsible for routing tenant traffic between internal and external networks?

A. Nova

B. Neutron

C. Keystone

D. Swift

Correct answer: B

Why B is right: Neutron provides networking: virtual networks, routers, DHCP, security groups and floating IPs.

Why the other options are wrong

A is wrong: Nova manages compute instances.

C is wrong: Keystone handles identity and authentication.

D is wrong: Swift is object storage.

Week 5

Week 5 · Question 1

A cloud provider guarantees 99% availability. A customer's application operates 12 hours per day. What is the maximum allowable downtime during a 30-day month without violating the SLA?

A. 3.6 hours

B. 6 hours

C. 7.2 hours

D. 9 hours

Correct answer: A

Why A is right: Scheduled time = 12 × 30 = 360 hours. Allowed downtime = 1% of 360 = 3.6 hours.

Why the other options are wrong

B is wrong: 6 hours is about 1.7% of 360 hours, which breaks the 99% guarantee.

C is wrong: 7.2 hours is 1% of 720 hours, which wrongly assumes the app runs 24 hours a day.

D is wrong: 9 hours is 2.5% of 360 hours, which breaks the 99% guarantee.

Week 5 · Question 2

A cloud server dynamically activates additional CPU cores to handle a sudden spike in user requests. What is the most likely impact on the server's power consumption?

A. Power consumption decreases because the workload is distributed.

B. Power consumption remains unchanged since CPU frequency is fixed.

C. Power consumption increases as additional CPU cores become active.

D. Power consumption depends only on the amount of RAM used.

Correct answer: C

Why C is right: Each active core draws power, so total consumption generally rises. DVFS can improve efficiency but does not remove this.

Why the other options are wrong

A is wrong: Spreading work across more cores does not lower total power; more cores are now running.

B is wrong: Fixed frequency does not matter; extra active cores still use extra power.

D is wrong: CPU activity affects power too, not only RAM.

Week 5 · Question 3

Which cloud characteristic most directly helps organizations avoid large upfront hardware investments?

A. Elastic scalability

B. Pay-as-you-go pricing

C. Fault tolerance

D. Multi-tenancy

Correct answer: B

Why B is right: With pay-as-you-go pricing, customers pay only for what they consume, so no big hardware purchase is needed up front.

Why the other options are wrong

A is wrong: Elasticity adjusts capacity to demand; it is not the pricing model that removes upfront cost.

C is wrong: Fault tolerance keeps services running through failures.

D is wrong: Multi-tenancy means many customers share resources; it is not the billing model.

Week 5 · Question 4

An HDFS cluster has a block size of 128 MB. A 700 MB input file is processed using MapReduce. Assuming one map task per HDFS block, how many map tasks are created?

A. 5

B. 6

C. 7

D. 8

Correct answer: B

Why B is right: 700 MB = 5 full blocks of 128 MB + 1 partial block of 60 MB = 6 blocks, so 6 map tasks.

Why the other options are wrong

A is wrong: 5 ignores the final partial block of 60 MB, which still needs a task.

C is wrong: 7 counts one block too many; 700 ÷ 128 is 5.47, rounded up to 6.

D is wrong: 8 counts two blocks too many.

Week 5 · Question 5

Which virtualization component is primarily responsible for creating and managing virtual machines?

A. Operating System Kernel

B. Hypervisor

C. BIOS

D. Device Driver

Correct answer: B

Why B is right: The hypervisor (Virtual Machine Monitor) creates, runs and manages VMs by abstracting the physical hardware.

Why the other options are wrong

A is wrong: The OS kernel manages processes on one OS; it does not create VMs.

C is wrong: The BIOS starts the hardware at boot.

D is wrong: A device driver lets the OS talk to one hardware device.

Week 5 · Question 6

Which MapReduce component determines which Reducer receives a particular key?

A. Combiner

B. Partitioner

C. RecordWriter

D. InputSplit

Correct answer: B

Why B is right: The Partitioner decides the destination reducer for each intermediate key; by default Hadoop uses a hash-based partitioner.

Why the other options are wrong

A is wrong: The Combiner pre-aggregates mapper output locally; it does not route keys.

C is wrong: RecordWriter writes the final output records.

D is wrong: InputSplit defines the chunk of input a mapper reads.

Week 5 · Question 7

Which MapReduce component is optional and may execute zero, one, or multiple times?

A. Mapper

B. Reducer

C. Combiner

D. RecordReader

Correct answer: C

Why C is right: The Combiner is an optional optimization and Hadoop does not guarantee it will run, so the algorithm must stay correct without it.

Why the other options are wrong

A is wrong: The Mapper is a required stage.

B is wrong: The Reducer is a required stage for jobs that aggregate.

D is wrong: The RecordReader always runs to turn input splits into key-value pairs.

Week 5 · Question 8

A cloud provider observes that many physical servers are running at only 15% CPU utilization. To reduce electricity costs, it migrates virtual machines onto fewer servers and powers off the idle hosts while ensuring that application performance remains within SLA limits. Which scheduling strategy is being applied?

A. Round-Robin Scheduling

B. Energy-Aware Scheduling

C. First-Come-First-Serve Scheduling

D. Fair Scheduling

Correct answer: B

Why B is right: Energy-aware scheduling consolidates workloads onto fewer machines so idle ones can be switched off while the required QoS is kept.

Why the other options are wrong

A is wrong: Round-robin hands out tasks in turn and spreads load, which does not power servers off.

C is wrong: FCFS only orders tasks by arrival time.

D is wrong: Fair scheduling divides resources evenly between users, not to save energy.

Week 5 · Question 9

Statement 1: Elasticity enables cloud users to increase or decrease resources according to workload demand. Statement 2: Elasticity helps organizations avoid paying for unused computing resources.

A. Both statements are true.

B. Both statements are false.

C. Statement 1 is true and Statement 2 is false.

D. Statement 1 is false and Statement 2 is true.

Correct answer: A

Why A is right: Elasticity scales resources up and down with demand, and releasing them when demand drops means no payment for idle infrastructure.

Why the other options are wrong

B is wrong: Both statements are true, so "both false" is wrong.

C is wrong: Statement 2 is also true, since released resources are no longer billed.

D is wrong: Statement 1 is also true; that is the definition of elasticity.

Week 5 · Question 10

Average demand = 75 units, Peak demand = 225 units. The cloud provider offers a utility premium of 2.8. Which statement is true?

A. Cloud cost < Owning cost

B. Cloud cost > Owning cost

C. Both costs are equal

D. None of the above

Correct answer: A

Why A is right: The break-even premium is peak ÷ average = 225 ÷ 75 = 3. Since 2.8 is below 3, the cloud is cheaper.

Why the other options are wrong

B is wrong: The cloud would be dearer only if the premium were above 3.

C is wrong: The costs would be equal only if the premium were exactly 3.

D is wrong: A correct statement exists, so "none" is wrong.

Week 6

Week 6 · Question 1

A database stores customer records in encrypted form. However, an administrator accidentally changes several account balances due to a software bug. Which security objective has failed?

A. Confidentiality only

B. Integrity only

C. Availability only

D. Confidentiality and Availability

Correct answer: B

Why B is right: Encryption protects confidentiality but cannot stop wrong modification by authorized users or software. Wrongly changed data is an integrity failure.

Why the other options are wrong

A is wrong: The data stayed encrypted, so confidentiality did not fail.

C is wrong: The data was still accessible, so availability did not fail.

D is wrong: Neither confidentiality nor availability was lost here.

Week 6 · Question 2

An employee logs into a fake VPN portal that perfectly imitates the company's official login page. Before any credentials are entered, which security property has already been compromised?

A. Confidentiality

B. Integrity

C. Availability

D. Authenticity

Correct answer: D

Why D is right: The fake portal falsely claims to be the real VPN service, so authenticity is broken before anything is disclosed.

Why the other options are wrong

A is wrong: No confidential data has been disclosed yet.

B is wrong: No data has been altered yet.

C is wrong: The real service is still up; nothing has been made unavailable.

Week 6 · Question 3

Which of the following attacks is NOT considered a passive attack?

A. Packet sniffing

B. Traffic analysis

C. Session hijacking

D. Network monitoring

Correct answer: C

Why C is right: Session hijacking takes over a live session by injecting or modifying packets, which is an active attack.

Why the other options are wrong

A is wrong: Packet sniffing only observes traffic, so it is passive.

B is wrong: Traffic analysis only observes traffic patterns, so it is passive.

D is wrong: Network monitoring only observes, so it is passive.

Week 6 · Question 4

Which of the following attacks cannot directly modify the contents of transmitted data?

A. Replay attack

B. Man-in-the-Middle attack

C. Eavesdropping

D. Masquerading

Correct answer: C

Why C is right: Eavesdropping only captures data in transit and does not alter or inject messages, so it is passive.

Why the other options are wrong

A is wrong: A replay attack resends captured messages, which is an active interference.

B is wrong: A man-in-the-middle can intercept and alter messages.

D is wrong: Masquerading means acting as someone else to send or alter traffic.

Week 6 · Question 5

Which of the following best describes a Denial-of-Service (DoS) attack?

A. An attacker secretly monitors network traffic without modifying it.

B. An attacker floods a server with excessive requests, making the service unavailable to legitimate users.

C. An attacker impersonates a legitimate user to gain unauthorized access.

D. An attacker modifies data during transmission between two communicating parties.

Correct answer: B

Why B is right: DoS is an active attack that overwhelms a service with requests so legitimate users cannot use it.

Why the other options are wrong

A is wrong: That describes passive monitoring or eavesdropping.

C is wrong: That describes masquerading.

D is wrong: That describes a man-in-the-middle or modification attack.

Week 6 · Question 6

Which of the following activities is least likely to be considered part of the reconnaissance phase?

A. Performing DNS enumeration to identify subdomains

B. Scanning the target to discover open ports and running services

C. Exploiting a buffer overflow vulnerability to gain administrator privileges

D. Searching public websites and social media for organizational information

Correct answer: C

Why C is right: Exploiting a vulnerability belongs to the attack (exploitation) phase, not information gathering.

Why the other options are wrong

A is wrong: DNS enumeration gathers information, so it is reconnaissance.

B is wrong: Port scanning gathers information about the target, so it is reconnaissance.

D is wrong: Searching public sources (OSINT) is reconnaissance.

Week 6 · Question 7

Statement I: Authentication determines whether a user is allowed to access a resource. Statement II: Authorization determines what actions an authenticated user is permitted to perform.

A. Statement I is TRUE and Statement II is FALSE.

B. Statement I is FALSE and Statement II is TRUE.

C. Both statements are TRUE.

D. Both statements are FALSE.

Correct answer: B

Why B is right: Authentication verifies who the user is; authorization decides what that user may do. Statement I describes authorization, so it is false; Statement II is true.

Why the other options are wrong

A is wrong: Statement II is true, and Statement I mixes up authentication with authorization.

C is wrong: Statement I is not true, since it describes authorization.

D is wrong: Statement II is true, so "both false" is wrong.

Week 6 · Question 8

A financial institution must install a custom Linux kernel module, configure firewall rules at the operating system level, and deploy proprietary database software. However, it does not want to purchase or maintain physical servers. Which cloud service model is most appropriate?

A. SaaS

B. PaaS

C. IaaS

D. FaaS

Correct answer: C

Why C is right: IaaS gives control of the OS and installed software while the provider handles the physical servers.

Why the other options are wrong

A is wrong: SaaS offers a finished application with no OS control.

B is wrong: PaaS manages the OS and runtime, so a custom kernel module is not possible.

D is wrong: FaaS runs small functions with no OS control.

Week 6 · Question 9

An organization discovers that backups of its confidential data are automatically replicated to another country without its knowledge. Which cloud security concern has been violated?

A. Data Location

B. Recovery

C. Virtualization

D. Resource Pooling

Correct answer: A

Why A is right: Automatic replication across regions can break legal or organizational rules about where sensitive data may be stored.

Why the other options are wrong

B is wrong: Recovery is about restoring data after failure; the issue is where the copies are.

C is wrong: Virtualization concerns isolation of VMs on shared hardware.

D is wrong: Resource pooling is the sharing of resources among customers.

Week 6 · Question 10

Which of the following cloud design practices is most effective in reducing vendor lock-in?

A. Using provider-specific APIs whenever possible

B. Developing applications using open standards and portable technologies

C. Storing data only in proprietary database formats

D. Relying exclusively on managed proprietary services

Correct answer: B

Why B is right: Open standards, containers and portable technologies reduce dependence on one provider and make migration easier.

Why the other options are wrong

A is wrong: Provider-specific APIs tie the application to that provider.

C is wrong: Proprietary formats make moving data hard.

D is wrong: Exclusive use of proprietary services increases lock-in.

Week 7

Week 7 · Question 1

Which of the following applications is least suitable for computation offloading to a remote cloud in Mobile Cloud Computing (MCC)?

A. Image rendering requiring intensive GPU computation

B. Large-scale machine learning inference

C. Emergency collision avoidance requiring millisecond response

D. Video transcoding

Correct answer: C

Why C is right: Safety-critical, real-time tasks need very low latency. Network delay and connectivity uncertainty make a remote cloud unsuitable; they belong on the device or on nearby edge/fog nodes.

Why the other options are wrong

A is wrong: Heavy GPU rendering is compute-intensive and tolerates some delay, so it suits offloading.

B is wrong: Large-scale ML inference is compute-intensive, so it suits offloading.

D is wrong: Video transcoding is heavy and not millisecond-critical, so it suits offloading.

Week 7 · Question 2

A cloud-based mobile application automatically allocates additional computing resources during a sudden surge in users and releases them when demand decreases. Which cloud characteristic is demonstrated?

A. Resource pooling

B. Rapid elasticity

C. Location independence

D. Broad network access

Correct answer: B

Why B is right: Rapid elasticity scales resources up or down automatically with workload.

Why the other options are wrong

A is wrong: Resource pooling is about sharing one pool among many customers.

C is wrong: Location independence means the user does not control or know the resource's location.

D is wrong: Broad network access means services are reachable over standard networks from many devices.

Week 7 · Question 3

Dynamic computation offloading primarily aims to optimize which combination of objectives?

A. Screen brightness and storage capacity

B. Execution time, energy consumption, and resource utilization

C. Camera quality and display refresh rate

D. Cloud subscription cost only

Correct answer: B

Why B is right: Dynamic offloading balances execution delay, battery use, CPU utilization and network conditions.

Why the other options are wrong

A is wrong: Screen brightness and storage are not what offloading decisions optimize.

C is wrong: Camera quality and refresh rate are hardware features, not offloading goals.

D is wrong: Cost matters, but it is not the only objective; time and energy count too.

Week 7 · Question 4

A mobile augmented reality application requires object recognition with an end-to-end response time below 20 ms. Which infrastructure is most appropriate?

A. A geographically distant public cloud data center

B. A nearby cloudlet connected through a high-speed local network

C. Local storage only

D. An offline mobile application

Correct answer: B

Why B is right: AR is highly latency-sensitive. A nearby cloudlet gives compute close to the user with minimal delay.

Why the other options are wrong

A is wrong: A distant data center adds too much network delay for 20 ms.

C is wrong: Storage alone provides no computing power for recognition.

D is wrong: An offline app lacks the compute for heavy object recognition.

Week 7 · Question 5

A healthcare application stores patient records in the cloud. Which security requirement is most critical to ensure that unauthorized users cannot access sensitive medical information?

A. Elasticity

B. Data confidentiality

C. Resource pooling

D. Load balancing

Correct answer: B

Why B is right: Confidentiality ensures only authorized people can read the data.

Why the other options are wrong

A is wrong: Elasticity concerns scaling, not access control.

C is wrong: Resource pooling is a sharing model, not a protection for data.

D is wrong: Load balancing spreads traffic; it does not restrict who sees records.

Week 7 · Question 6

Which limitation of traditional cloud computing is primarily addressed by fog computing?

A. Limited virtualization support

B. High network latency caused by distant cloud data centers

C. Lack of cloud storage

D. Insufficient computing power in cloud servers

Correct answer: B

Why B is right: Fog places compute closer to end devices, cutting the delay of sending data to a distant data center.

Why the other options are wrong

A is wrong: Clouds already have strong virtualization.

C is wrong: Clouds offer plenty of storage.

D is wrong: Cloud servers have abundant computing power; distance is the problem.

Week 7 · Question 7

Which statement about fog nodes is FALSE?

A. They may perform filtering and aggregation of sensor data.

B. They are usually located close to IoT devices.

C. They permanently store all enterprise data.

D. They reduce communication with distant cloud servers.

Correct answer: C

Why C is right: Fog nodes do temporary processing and short-term storage; long-term storage is mostly in cloud data centers. So this statement is false.

Why the other options are wrong

A is wrong: True: fog nodes filter and aggregate sensor data.

B is wrong: True: fog nodes sit near IoT devices.

D is wrong: True: by processing locally, fog nodes cut traffic to the cloud.

Week 7 · Question 8

Statement 1: Geospatial Cloud Computing enables parallel processing of massive satellite imagery by distributing computation across multiple cloud nodes. Statement 2: Distributed storage and parallel processing frameworks reduce execution time for large geospatial datasets compared with a single desktop workstation.

A. Both Statement 1 and Statement 2 are true, and Statement 2 is the correct explanation of Statement 1

B. Both Statement 1 and Statement 2 are true, but Statement 2 is not the correct explanation of Statement 1

C. Statement 1 is true, but Statement 2 is false

D. Statement 1 is false, but Statement 2 is true

Correct answer: A

Why A is right: Both are true, and distributed storage with parallel computation is exactly why clouds process huge geospatial datasets efficiently.

Why the other options are wrong

B is wrong: Statement 2 does explain Statement 1, so "not the explanation" is wrong.

C is wrong: Statement 2 is true, so it cannot be false.

D is wrong: Statement 1 is true, so it cannot be false.

Week 7 · Question 9

A cloud platform divides a large raster image into tiles for parallel processing. Which challenge must be carefully addressed to ensure accurate results?

A. Screen resolution of client devices

B. Boundary effects between adjacent image tiles

C. Keyboard input latency

D. Number of GIS users

Correct answer: B

Why B is right: Filtering, convolution and terrain analysis need neighbouring pixels. Tiles can cause errors at their edges unless overlap or edge handling is used.

Why the other options are wrong

A is wrong: Client screen resolution does not affect the processing result.

C is wrong: Keyboard latency is unrelated to tile processing.

D is wrong: The number of users affects load, not accuracy of tiled results.

Week 7 · Question 10

A wearable ECG sensor continuously streams a patient's heart signals. Which architecture is most appropriate to detect cardiac abnormalities within a few milliseconds?

A. IoT sensors → Remote Cloud → Physician

B. IoT sensors → Fog/Edge Node → Cloud

C. IoT sensors → Local Storage → Weekly Upload

D. IoT sensors → FTP Server → Desktop GIS

Correct answer: B

Why B is right: Delay-sensitive healthcare needs immediate processing at a nearby fog/edge node; the cloud handles long-term storage, analysis and model training.

Why the other options are wrong

A is wrong: A remote cloud adds too much delay for millisecond detection.

C is wrong: A weekly upload is far too slow for detection.

D is wrong: An FTP server and desktop GIS are not real-time health-monitoring tools.

Week 8

Week 8 · Question 1

Which command is used to remove a Docker image that is no longer required?

A. docker rm

B. docker rmi

C. docker delete

D. docker remove

Correct answer: B

Why B is right: docker rmi removes one or more images from the local system.

Why the other options are wrong

A is wrong: docker rm removes containers, not images.

C is wrong: docker delete is not a Docker command for this.

D is wrong: docker remove is not the correct command for images.

Week 8 · Question 2

Which Docker command is primarily used to list all available Docker networks?

A. docker network ls

B. docker network show

C. docker net ps

D. docker networks

Correct answer: A

Why A is right: docker network ls lists the networks with ID, name, driver and scope.

Why the other options are wrong

B is wrong: docker network show is not a valid listing command.

C is wrong: docker net ps is not a valid command.

D is wrong: docker networks is not a valid command.

Week 8 · Question 3

Which of the following is the most accurate comparison between VMs and containers?

A. VMs share the host kernel, while containers have separate kernels

B. Containers generally share the host kernel, while VMs virtualize hardware and run separate guest OS kernels

C. Both always require separate guest operating systems

D. Containers require more resources than VMs because they include a complete OS

Correct answer: B

Why B is right: Containers isolate applications at the OS level and share the host kernel; a VM runs its own guest OS on virtualized hardware.

Why the other options are wrong

A is wrong: This swaps the two: VMs have their own kernels and containers share.

C is wrong: Containers do not need a separate guest OS.

D is wrong: Containers are generally lighter than VMs, not heavier.

Week 8 · Question 4

What is the primary role of etcd in a Kubernetes cluster?

A. Running application containers

B. Storing Kubernetes cluster state and configuration data

C. Scheduling Pods onto Nodes

D. Providing container images

Correct answer: B

Why B is right: etcd is a distributed key-value store holding cluster state, configuration and metadata for the control plane.

Why the other options are wrong

A is wrong: Containers run on worker nodes through the container runtime.

C is wrong: Scheduling is done by the kube-scheduler.

D is wrong: Images come from a container registry.

Week 8 · Question 5

Which of the following is an example of a Type 2 (hosted) hypervisor?

A. VMware ESXi

B. Xen

C. Oracle VirtualBox

D. Microsoft Hyper-V Server

Correct answer: C

Why C is right: VirtualBox runs as an application on top of an existing host OS, which makes it Type 2.

Why the other options are wrong

A is wrong: ESXi runs directly on hardware (Type 1).

B is wrong: Xen runs directly on hardware (Type 1).

D is wrong: Hyper-V Server runs directly on hardware (Type 1).

Week 8 · Question 6

A cloud data center has 100 physical servers, but only 40 are heavily utilized while the remaining servers operate at very low utilization. Which strategy would MOST effectively reduce energy consumption?

A. Distribute VMs equally across all 100 servers

B. Consolidate VMs onto fewer servers and switch idle servers to a low-power state

C. Increase CPU frequency on all servers

D. Duplicate every VM across multiple servers

Correct answer: B

Why B is right: Consolidating VMs onto fewer servers lets the idle ones sleep or power off, cutting energy use.

Why the other options are wrong

A is wrong: Spreading VMs over all servers keeps every server powered on.

C is wrong: Higher CPU frequency raises power use.

D is wrong: Duplicating VMs adds load and power use.

Week 8 · Question 7

In a Sensor Cloud environment, why is a middleware layer often required between Wireless Sensor Networks (WSNs) and cloud services?

A. To increase the physical size of sensor nodes

B. To abstract heterogeneous sensors and provide cloud-accessible sensor resources

C. To eliminate all communication between sensors

D. To replace the sensing function of WSN nodes

Correct answer: B

Why B is right: Middleware hides differences between resource-constrained sensors and handles discovery, data collection, protocol translation and service management for the cloud.

Why the other options are wrong

A is wrong: Middleware is software; it does not change physical size.

C is wrong: Sensors still need to communicate; middleware does not remove that.

D is wrong: Sensors still do the sensing; middleware does not replace them.

Week 8 · Question 8

Which architecture is generally BEST suited for applications requiring both low latency and large-scale cloud analytics?

A. IoT Device → Cloud only

B. IoT Device → Edge/Fog → Cloud

C. IoT Device → Database only

D. IoT Device → Hypervisor → Cloud

Correct answer: B

Why B is right: Edge/fog nodes handle latency-sensitive work locally, while the cloud provides large-scale storage, history analysis and heavy computation.

Why the other options are wrong

A is wrong: Cloud-only has higher latency from distance.

C is wrong: A database alone provides neither local processing nor large-scale analytics.

D is wrong: A hypervisor is virtualization software, not a latency-reducing layer.

Week 8 · Question 9

An IoT application sends sensitive sensor data through public networks to a cloud platform. Which mechanism is MOST important for protecting the data during transmission?

A. Data encryption

B. Increasing CPU frequency

C. Increasing sensor sampling rate

D. VM consolidation

Correct answer: A

Why A is right: Encryption (for example TLS) stops unauthorized parties from reading or modifying data in transit.

Why the other options are wrong

B is wrong: CPU frequency affects speed, not data protection.

C is wrong: Sampling rate changes how much data is collected, not its security.

D is wrong: VM consolidation saves energy; it does not protect data.

Week 8 · Question 10

Statement 1: Processing latency-sensitive IoT workloads at an edge node can reduce the communication delay associated with a distant cloud. Statement 2: Edge processing eliminates all network latency because no communication is required.

A. Statement 1 is True and Statement 2 is False

B. Statement 2 is True and Statement 1 is False

C. Both statements are True

D. Both statements are False

Correct answer: A

Why A is right: Edge moves processing closer to the data and reduces delay, but some communication latency remains, so Statement 2 is false.

Why the other options are wrong

B is wrong: Statement 2 is false, since some latency always remains.

C is wrong: Statement 2 is false, so "both true" is wrong.

D is wrong: Statement 1 is true, so "both false" is wrong.

Week 9

Week 9 · Question 1

A factory requires machine-level anomaly detection within milliseconds, but it also needs to aggregate data from multiple machines for plant-wide analysis before sending summaries to the cloud. Which placement BEST matches the requirements?

A. All processing in the centralized cloud

B. Machine-level processing at the edge and plant-level processing at fog nodes

C. All processing directly on individual sensors

D. Plant-wide processing exclusively at the cloud

Correct answer: B

Why B is right: Edge handles the millisecond-level work next to each machine; fog nodes aggregate across machines and send summaries to the cloud.

Why the other options are wrong

A is wrong: Cloud-only is too far away for millisecond detection.

C is wrong: Sensors alone cannot combine data across machines for plant-wide analysis.

D is wrong: The brief wants summaries sent to the cloud after plant-level aggregation, not raw plant-wide processing there.

Week 9 · Question 2

A smart-city traffic system receives video streams from hundreds of cameras. Immediate object detection is performed near each camera, regional congestion patterns are analyzed at roadside micro-data centers, and long-term traffic trends are analyzed in the cloud. Which architecture BEST represents this system?

A. Cloud-only architecture

B. Fog-only architecture

C. Edge–Fog–Cloud architecture

D. Edge-only architecture

Correct answer: C

Why C is right: It uses edge for immediate detection, fog for regional analysis and the cloud for long-term trends: all three layers.

Why the other options are wrong

A is wrong: The system clearly uses processing outside the cloud.

B is wrong: It also uses edge and cloud, not only fog.

D is wrong: It also uses fog and cloud, not only edge.

Week 9 · Question 3

An IoT system performs local filtering before forwarding data to the cloud. Which ordering BEST represents the typical amount of raw data transmitted toward centralized cloud infrastructure, from lowest to highest?

A. Edge processing → Fog processing → Cloud-only processing

B. Cloud-only processing → Fog processing → Edge processing

C. Fog processing → Cloud-only processing → Edge processing

D. Edge processing → Cloud-only processing → Fog processing

Correct answer: A

Why A is right: Edge filters data first (least sent), fog aggregates regionally (more), and cloud-only sends all raw data (most).

Why the other options are wrong

B is wrong: This order is reversed: cloud-only sends the most.

C is wrong: Edge filtering sends less than cloud-only, so edge cannot come last.

D is wrong: Fog aggregation sends less than cloud-only, so fog cannot come last.

Week 9 · Question 4

A fog node is running out of memory, but the infrastructure allows additional RAM and CPU resources to be assigned to that same node. Which scalability approach is being used?

A. Horizontal scalability

B. Temporal scalability

C. Geographic scalability

D. Vertical scalability

Correct answer: D

Why D is right: Vertical scalability (scale-up) adds resources such as CPU and RAM to an existing node.

Why the other options are wrong

A is wrong: Horizontal scalability adds more nodes instead.

B is wrong: Temporal scalability is not the term for adding resources to a node.

C is wrong: Geographic scalability is about spreading across locations.

Week 9 · Question 5

Which statement BEST describes the division of responsibilities in a hierarchical Edge–Fog–Cloud architecture?

A. Edge handles global optimization, while Cloud handles sensor-level decisions

B. Edge handles immediate local actions, Fog coordinates regional resources, and Cloud performs global orchestration

C. All three layers perform exactly the same tasks

D. Only the Cloud is allowed to make decisions

Correct answer: B

Why B is right: Decisions are split by proximity and scope: edge acts locally, fog coordinates regionally, and the cloud sets global policy and long-term analytics.

Why the other options are wrong

A is wrong: This swaps the roles of edge and cloud.

C is wrong: The layers exist because they do different jobs.

D is wrong: Edge and fog also make decisions.

Week 9 · Question 6

A video analytics application requires its first-stage processing to occur within 10 ms of the camera, while its model-training component requires substantial computational resources. Which placement is MOST appropriate?

A. Both components on a distant cloud server

B. Both components on the smallest IoT device

C. First-stage processing at the edge and model training in the cloud

D. First-stage processing only in the cloud and training at the edge

Correct answer: C

Why C is right: The 10 ms stage must be next to the camera (edge); heavy model training belongs where compute is plentiful (cloud).

Why the other options are wrong

A is wrong: A distant cloud cannot meet 10 ms for the first stage.

B is wrong: The smallest IoT device lacks the power for model training.

D is wrong: This reverses the placement: the cloud is too far for 10 ms, and the edge is too weak for training.

Week 9 · Question 7

A mobile device is moving rapidly between coverage areas of different edge servers while executing an offloaded task. Which issue becomes MOST critical?

A. Static CPU allocation

B. Service migration and task continuity during handover

C. Cloud storage capacity

D. VM image size only

Correct answer: B

Why B is right: As the device changes coverage areas, the task may need migrating or reassigning with minimal interruption and overhead.

Why the other options are wrong

A is wrong: CPU allocation is not the main problem when moving between servers.

C is wrong: Cloud storage is not what breaks during movement.

D is wrong: Image size is only one factor; continuity during handover is the key issue.

Week 9 · Question 8

A cloud application requires additional resources because its primary CSP has insufficient capacity. A broker identifies a suitable secondary CSP and negotiates the required service conditions. Which cloud-federation function is being performed?

A. Resource federation and inter-provider service provisioning

B. VM deletion within a single CSP

C. Local container orchestration

D. Hardware virtualization

Correct answer: A

Why A is right: A federation broker discovers resources across several providers and coordinates provisioning between them.

Why the other options are wrong

B is wrong: Nothing is being deleted, and a second provider is involved.

C is wrong: Orchestrating containers happens inside one environment, not across providers.

D is wrong: Hardware virtualization is a single-host technology, not a provider-to-provider function.

Week 9 · Question 9

Why is trust management particularly important in a cloud federation?

A. Resources and services may be obtained from independent providers with different security and reliability policies

B. All CSPs are automatically controlled by the same organization

C. Federation eliminates authentication requirements

D. Trust is required only for physical servers

Correct answer: A

Why A is right: Federated CSPs are independent, with different security, compliance and reliability, so trust must be built and verified.

Why the other options are wrong

B is wrong: CSPs in a federation are independent, not controlled by one organization.

C is wrong: Federation does not remove authentication; it makes it more important.

D is wrong: Trust applies to services, data and providers, not only physical servers.

Week 9 · Question 10

Which is the PRIMARY advantage of a Hybrid Cloud Federation for an organization handling sensitive workloads?

A. It eliminates the need for security controls

B. It forces all workloads into the public cloud

C. It combines private-cloud control for sensitive workloads with public-cloud scalability

D. It prevents workload migration between environments

Correct answer: C

Why C is right: Sensitive workloads stay in private infrastructure while less-sensitive or burst workloads use public-cloud capacity.

Why the other options are wrong

A is wrong: Security controls are still needed.

B is wrong: Hybrid does not force everything into the public cloud.

D is wrong: Hybrid allows workloads to move between environments.

Week 10

Week 10 · Question 1

During live migration of a VM, which condition is most likely to increase the total migration time?

A. Low rate of memory-page modification by the VM

B. High rate of memory-page modification during migration

C. Increasing the number of CPU cores on the destination host

D. Reducing the VM memory size

Correct answer: B

Why B is right: Pages modified during copying become dirty and must be sent again, so a high modification rate means more data and a longer migration.

Why the other options are wrong

A is wrong: Few modified pages means little re-copying, so migration is faster.

C is wrong: More destination cores do not reduce the data to be sent.

D is wrong: A smaller memory means less data to transfer.

Week 10 · Question 2

Which live VM migration technique starts the VM on the destination first and then fetches memory pages from the source on demand as they are accessed?

A. Cold migration

B. Pre-copy migration

C. Post-copy migration

D. Offline image transfer

Correct answer: C

Why C is right: Post-copy resumes the VM at the destination right away and pulls pages from the source on demand (page faults) or by background pushing.

Why the other options are wrong

A is wrong: Cold migration stops the VM first and moves it.

B is wrong: Pre-copy transfers memory before switching over.

D is wrong: Offline transfer also needs the VM stopped.

Week 10 · Question 3

Which technique is most likely to reduce the total data transferred (and hence the energy used) during pre-copy live migration?

A. Increasing the memory-dirtying rate of the workload

B. Compressing and deduplicating memory pages before transfer

C. Adding more iterative pre-copy rounds indefinitely

D. Disabling dirty-page tracking

Correct answer: B

Why B is right: Compression and deduplication shrink the data sent, saving time and energy.

Why the other options are wrong

A is wrong: A higher dirtying rate means more retransmission.

C is wrong: Unlimited extra rounds waste bandwidth.

D is wrong: Without dirty-page tracking, pre-copy cannot be correct.

Week 10 · Question 4

Which resource management strategy is most important to prevent overload when many edge devices send data to nearby fog nodes?

A. Increasing storage capacity at a single fog node

B. Static task allocation

C. Load balancing across fog nodes

D. Sending all data directly to the cloud

Correct answer: C

Why C is right: Load balancing spreads tasks over several fog nodes, avoiding congestion and keeping latency low.

Why the other options are wrong

A is wrong: More storage on one node does not fix its processing overload.

B is wrong: Static allocation cannot adapt to changing load.

D is wrong: Sending everything to the cloud adds delay and bandwidth use, defeating fog's purpose.

Week 10 · Question 5

A Kubernetes cluster has several worker nodes. A newly created Pod must be assigned to an appropriate node based on available resources and scheduling constraints. Which Kubernetes component is primarily responsible for this decision?

A. kubelet

B. kube-scheduler

C. kube-proxy

D. Container registry

Correct answer: B

Why B is right: The kube-scheduler picks a node by looking at resource needs, availability, affinity rules and constraints.

Why the other options are wrong

A is wrong: The kubelet manages Pods on a node after the choice is made.

C is wrong: kube-proxy handles networking rules for services.

D is wrong: A container registry stores images.

Week 10 · Question 6

Consider the following Docker workflow: Dockerfile → docker build → _____ → docker run. Which entity is produced by the docker build operation?

A. Docker volume

B. Docker image

C. Running container

D. Kubernetes Pod

Correct answer: B

Why B is right: docker build reads the Dockerfile and creates an image; docker run later creates a container from it.

Why the other options are wrong

A is wrong: A volume is storage created separately.

C is wrong: A running container comes from docker run, not docker build.

D is wrong: A Pod is a Kubernetes object, not a Docker build output.

Week 10 · Question 7

Which of the following correctly describes the roles of Linux namespaces and cgroups in containers?

A. Namespaces provide resource quotas, while cgroups provide process isolation.

B. Namespaces provide isolation of processes and system resources, while cgroups control resource usage.

C. Both namespaces and cgroups are used only to create Docker images.

D. Neither is required for container isolation.

Correct answer: B

Why B is right: Namespaces give separate views of process IDs, networking and mounts; cgroups limit and monitor CPU, memory and other resources.

Why the other options are wrong

A is wrong: This swaps the roles of namespaces and cgroups.

C is wrong: They run containers; they do not build images.

D is wrong: Both are core to container isolation and resource control.

Week 10 · Question 8

A VM has 32 GB of memory and the migration link provides 16 MB/sec of usable throughput. For a non-live (cold) migration where the entire memory is transferred once, what is the approximate transfer time? (Use 1 GB = 2^30 bytes, 1 MB = 2^20 bytes.)

A. About 15 minutes

B. About 34 minutes

C. About 1 hour

D. About 2 hours

Correct answer: B

Why B is right: Time = (32 × 1024 MB) ÷ 16 MB/s = 2048 s, about 34 minutes.

Why the other options are wrong

A is wrong: 15 minutes (900 s) would need about 36 MB/s.

C is wrong: 1 hour (3600 s) would need about 9 MB/s.

D is wrong: 2 hours (7200 s) would need about 4.6 MB/s.

Week 10 · Question 9

A database container is deleted accidentally. The database files were stored using a Docker named volume rather than only inside the container's writable layer. What happens to the data?

A. The data is automatically deleted with the container.

B. The data remains available in the named volume.

C. The data is converted into a Docker image.

D. The data can only be recovered by restarting the deleted container.

Correct answer: B

Why B is right: A volume has a lifecycle independent of containers, so deleting the container does not delete its data.

Why the other options are wrong

A is wrong: Data in the container's writable layer is lost, but named volumes persist.

C is wrong: Volume data is never turned into an image.

D is wrong: The container is gone, but the volume can be attached to a new one.

Week 10 · Question 10

Which of the following statements about containers and virtual machines is CORRECT?

A. Containers normally require a separate complete guest operating system for every application.

B. Virtual machines and containers always provide identical levels of isolation.

C. Containers generally share the host operating system kernel, whereas VMs virtualize a complete machine environment.

D. Containers cannot be deployed on virtual machines.

Correct answer: C

Why C is right: Containers share the host kernel, so they are lighter and start faster; a VM virtualizes hardware and runs its own guest OS.

Why the other options are wrong

A is wrong: Containers do not need a guest OS each.

B is wrong: VMs generally isolate more strongly than containers.

D is wrong: Containers can run inside VMs.

Week 11

Week 11 · Question 1

Which capability is the defining feature of Dew computing compared to conventional cloud computing?

A. It requires a permanent high-bandwidth connection to the cloud

B. It provides local independence, letting a device store and process data offline and sync with the cloud when connectivity returns

C. It eliminates the need for any cloud component

D. It performs only data storage and never computation

Correct answer: B

Why B is right: Dew computing gives local independence: devices keep working offline and reconcile with the cloud once reconnected.

Why the other options are wrong

A is wrong: This is the opposite: dew computing works without a permanent connection.

C is wrong: Dew computing complements the cloud; it does not replace it.

D is wrong: Dew devices do both storage and computation.

Week 11 · Question 2

A weather-monitoring device installed in a remote mountain region must continue recording and processing measurements for several hours even when the Internet connection is unavailable. Once connectivity is restored, the accumulated data is synchronized with a cloud server. Which computing paradigm best matches this requirement?

A. Edge computing

B. Fog computing

C. Dew computing

D. Centralized cloud computing

Correct answer: C

Why C is right: Dew computing stresses autonomous operation at or near the device, with later synchronization to the cloud.

Why the other options are wrong

A is wrong: Edge computing processes near the source but does not stress offline operation with later sync.

B is wrong: Fog computing uses nearby intermediate nodes that still need network connectivity.

D is wrong: A centralized cloud cannot work while the Internet connection is down.

Week 11 · Question 3

Two users modify the same record independently while their devices are offline. When both devices reconnect, the cloud detects that the two versions conflict. Which mechanism is most directly required to handle this situation?

A. CPU scheduling

B. Conflict detection and resolution during synchronization

C. Container image creation

D. Load balancing

Correct answer: B

Why B is right: Offline edits create conflicting versions, so the system must detect the conflict and apply a resolution policy to restore a consistent state.

Why the other options are wrong

A is wrong: CPU scheduling decides which process runs, not which data version wins.

C is wrong: Creating container images is unrelated to data conflicts.

D is wrong: Load balancing spreads requests; it does not merge versions.

Week 11 · Question 4

Allowing Dew devices to operate offline and modify data independently most directly introduces which challenge?

A. Increased CPU virtualization overhead

B. The need to detect and resolve data conflicts when devices reconnect and synchronize

C. Permanent loss of all local data

D. Inability to ever reach the cloud

Correct answer: B

Why B is right: Independent offline changes let copies diverge, so consistency must be restored at sync: the autonomy-versus-consistency trade-off.

Why the other options are wrong

A is wrong: Offline operation does not add virtualization overhead.

C is wrong: Local data is kept and synchronized later, not lost.

D is wrong: Devices can reach the cloud when connectivity returns.

Week 11 · Question 5

A serverless function has not been invoked for a long period. When the next request arrives, the platform requires additional time to initialize the execution environment. This delay is known as:

A. Warm start

B. Cold start

C. Cache miss

D. Bootstrapping failure

Correct answer: B

Why B is right: A cold start happens when the platform has to create or initialize a new execution environment before running the request, which adds latency.

Why the other options are wrong

A is wrong: A warm start reuses an environment that is already ready, so there is no extra delay.

C is wrong: A cache miss concerns missing cached data, not environment setup.

D is wrong: Bootstrapping failure is not the term for this delay, and nothing has failed.

Week 11 · Question 6

Why are serverless functions typically designed to be stateless?

A. To reduce electricity consumption

B. Because local storage is unlimited

C. To allow automatic scaling and parallel execution without dependency conflicts

D. To avoid using APIs

Correct answer: C

Why C is right: Stateless invocations are independent, so the platform can run many instances in parallel and scale horizontally without coordination issues.

Why the other options are wrong

A is wrong: Saving electricity is not the design reason.

B is wrong: Local storage in functions is limited and temporary.

D is wrong: Serverless functions use APIs freely.

Week 11 · Question 7

A team wants to run a batch job that takes about 45 minutes to complete in a single AWS Lambda invocation. Why is this NOT feasible?

A. Lambda has no internet access

B. Lambda enforces a maximum execution timeout of 15 minutes per invocation

C. Lambda cannot process batch data

D. Lambda functions must always be stateful

Correct answer: B

Why B is right: A single Lambda invocation is capped at 15 minutes (900 seconds). A 45-minute job must be split up or run on containers, AWS Batch or EC2.

Why the other options are wrong

A is wrong: Lambda can have internet access.

C is wrong: Lambda can process batch data in smaller pieces.

D is wrong: Lambda functions are designed to be stateless, not stateful.

Week 11 · Question 8

A serverless application receives 1,000 independent requests simultaneously. The provider can create multiple function instances to process these requests concurrently. Which property of serverless computing makes this possible?

A. Stateful execution

B. Horizontal scalability of independent function invocations

C. Fixed allocation of one VM per user

D. Permanent local storage

Correct answer: B

Why B is right: Independent invocations let the platform automatically add function instances as demand rises, so many run in parallel.

Why the other options are wrong

A is wrong: Stateful execution would make parallel copies depend on each other.

C is wrong: A fixed VM per user does not scale to 1,000 simultaneous requests.

D is wrong: Permanent local storage is not a serverless property and does not enable concurrency.

Week 11 · Question 9

A research group wants to execute a scientific computation that requires 40 minutes of uninterrupted execution for each task. The chosen FaaS platform imposes a maximum execution time of 15 minutes per invocation. What is the primary problem with using this platform directly?

A. Functions cannot communicate with databases.

B. The execution time exceeds the platform's maximum invocation duration.

C. FaaS cannot process numerical calculations.

D. Serverless functions cannot execute in parallel.

Correct answer: B

Why B is right: Each invocation has a time limit, and a 40-minute task cannot finish within a 15-minute limit.

Why the other options are wrong

A is wrong: Functions can communicate with databases.

C is wrong: FaaS can run numerical calculations.

D is wrong: Serverless functions can run in parallel.

Week 11 · Question 10

A company needs a managed cloud platform specifically designed for large-scale analytical queries over structured datasets, using distributed processing rather than relying on a single database server. Which service is most appropriate?

A. Amazon Redshift

B. Amazon EC2

C. AWS Lambda

D. Amazon Route 53

Correct answer: A

Why A is right: Redshift is a managed cloud data warehouse built for large-scale analytics, using distributed, column-oriented query processing.

Why the other options are wrong

B is wrong: EC2 gives raw virtual machines, not a managed analytics platform.

C is wrong: Lambda runs short functions, not large analytical queries.

D is wrong: Route 53 is a DNS service.

Week 12

Week 12 · Question 1

Mobile Cloud Computing (MCC) benefits mobile devices by: 1. Offloading compute-intensive tasks to the cloud 2. Extending effective battery life by reducing on-device processing 3. Guaranteeing zero network latency 4. Providing access to large-scale storage

A. 1, 2, and 4 only

B. 1 and 3 only

C. 2 and 3 only

D. All of the above

Correct answer: A

Why A is right: MCC offloads heavy work (1), which saves battery (2), and gives access to scalable storage (4). It cannot guarantee zero latency (3), since wireless communication always adds delay.

Why the other options are wrong

B is wrong: This includes the false statement 3 and leaves out 2 and 4.

C is wrong: This includes the false statement 3.

D is wrong: This includes the false statement 3, so "all" is wrong.

Week 12 · Question 2

A city planning department stores road networks, building locations, GPS coordinates, and satellite images in a cloud-based system for analysis. This is primarily an example of:

A. Mobile Cloud Computing

B. Spatial Cloud Computing

C. Serverless Computing

D. Batch Computing

Correct answer: B

Why B is right: Spatial cloud computing manages and processes geospatial data: coordinates, maps, GIS layers and satellite imagery.

Why the other options are wrong

A is wrong: Mobile cloud computing is about mobile devices using the cloud.

C is wrong: Serverless computing is about running functions without managing servers.

D is wrong: Batch computing is about processing jobs in groups, not about spatial data.

Week 12 · Question 3

Which task is MOST likely to rely on Spatial Cloud Computing?

A. Sorting a company's payroll spreadsheet

B. Real-time flood-risk mapping using terrain and rainfall data across a region

C. Compiling source code

D. Streaming a music playlist

Correct answer: B

Why B is right: Regional flood-risk mapping combines terrain, rainfall and location data, which is large-scale geospatial analytics.

Why the other options are wrong

A is wrong: A payroll spreadsheet has no spatial data.

C is wrong: Compiling code has no spatial data.

D is wrong: Streaming music has no spatial data.

Week 12 · Question 4

In Cyber-Physical Systems (CPS), real-time feedback between sensors and actuators is essential for:

A. Data archiving

B. Closed-loop control

C. Static data analysis

D. Batch processing

Correct answer: B

Why B is right: CPS joins sensing, computation and actuation: sensors read, computation decides, actuators act, forming a closed control loop.

Why the other options are wrong

A is wrong: Archiving stores data and needs no real-time feedback.

C is wrong: Static analysis works on fixed data, not live feedback.

D is wrong: Batch processing works on groups of data later, not in real time.

Week 12 · Question 5

A GPS-enabled taxi periodically records its location as: (x1,y1,t1),(x2,y2,t2),…,(xn,yn,tn). What does this sequence represent?

A. Spatial trajectory

B. Spatial database

C. Road topology

D. Spatial index

Correct answer: A

Why A is right: A spatial trajectory is a time-ordered sequence of positions showing a moving object's path.

Why the other options are wrong

B is wrong: A spatial database is a system that stores spatial data, not one path.

C is wrong: Road topology describes how roads connect, not a taxi's movement.

D is wrong: A spatial index is a structure for fast spatial search.

Week 12 · Question 6

_____ refers to outsourcing location-based sensing or data-collection tasks to a crowd of mobile users who gather data at specific places and times.

A. Spatial trajectory

B. Spatial crowdsourcing

C. Time series

D. Road network graph

Correct answer: B

Why B is right: Spatial crowdsourcing assigns location- and time-specific sensing tasks to mobile participants, such as reporting road conditions at given spots.

Why the other options are wrong

A is wrong: A spatial trajectory is a moving object's time-ordered path, not a task-assignment method.

C is wrong: A time series is data ordered in time, not a way to outsource tasks.

D is wrong: A road network graph models roads, not crowd tasks.

Week 12 · Question 7

In V2V communication, the primary objective is:

A. Providing internet access to vehicles

B. Exchanging safety-critical information such as speed, position, and braking

C. Supporting infotainment systems

D. Uploading vehicle data to cloud storage

Correct answer: B

Why B is right: V2V sends direct vehicle-to-vehicle safety messages (speed, heading, braking, lane changes) to prevent collisions.

Why the other options are wrong

A is wrong: V2V is not meant to provide internet access.

C is wrong: V2V is not meant for entertainment.

D is wrong: Uploading to the cloud is not V2V's primary goal; it is direct vehicle-to-vehicle messaging.

Week 12 · Question 8

A connected-vehicle system where onboard sensors capture data, roadside units preprocess and react locally, and the cloud performs fleet-wide long-term analytics is BEST described as:

A. Cloud-only computing

B. Edge computing with no cloud

C. An integrated Edge–Fog–Cloud architecture

D. A single centralized server

Correct answer: C

Why C is right: Onboard sensors (edge), roadside units (fog) and the cloud for long-term analytics spread computation over three layers.

Why the other options are wrong

A is wrong: The system also processes locally, so it is not cloud-only.

B is wrong: The cloud does fleet-wide analytics, so the cloud is used.

D is wrong: Computation is distributed, not on one central server.

Week 12 · Question 9

A major benefit of real-time mobility analytics in urban transport systems is:

A. Eliminating the need for traffic laws

B. Dynamic traffic signal optimization based on live vehicle density

C. Reducing vehicle manufacturing costs

D. Preventing all road accidents

Correct answer: B

Why B is right: Live data from sensors, cameras and GPS lets signals and routes adapt dynamically, reducing congestion and delays.

Why the other options are wrong

A is wrong: Analytics does not remove the need for traffic laws.

C is wrong: Analytics does not change manufacturing costs.

D is wrong: Analytics can reduce accidents but cannot prevent all of them.

Week 12 · Question 10

A multimedia application needs to deliver extremely high-resolution video and augmented-reality content to mobile users. Which 5G service category is most directly suited to this requirement?

A. eMBB

B. mMTC

C. URLLC

D. NFC

Correct answer: A

Why A is right: Enhanced Mobile Broadband (eMBB) provides very high data rates and bandwidth efficiency, suited to high-resolution video and AR/VR.

Why the other options are wrong

B is wrong: mMTC serves huge numbers of low-power IoT devices, not high data rates.

C is wrong: URLLC targets ultra-reliable low latency, not maximum bandwidth.

D is wrong: NFC is a short-range contactless technology, not a 5G service category.
